This is not from 2013. We believe it to be the 2014 Test. BEngTuks Team.

Physics Department
Fisika Departement

University of Pretoria

Semester Test 1 | *Semestertoets 1*

FSK116

15 March 2013 / *15 Maart ~~2013~~*

Surname and initials *Van en voorletters*	MEMO	Student number *Studentenommer*	1400000
Signature *Handtekening*			
Time *Tyd*	2½ hours *2½ ure*	Max. Total *Maks. Totaal*	65 Marks Max *65 punte Maks*
Internal Examiner: *Interne Eksaminator*	Dr J M Nel	External Examiner *Eksterne Eksaminator*	Mr R Q Odendaal

• Answer all questions. • Cleary show all calculations. Do not just write down the answer. • Clearly state all assumptions, rules and laws used. • Where applicable draw the free-body diagrams. • Demonstrate clearly that you understand the physics that you use to answer the question. This means that a numerically correct answer will not necessarily be awarded full marks. • Programmable calculators may NOT be used. • No textbooks allowed. • A formula sheet is attached. • All test and examination regulations of the University of Pretoria are applicable.	• *Beantwoord alle vrae.* • *Wys al jou berekeninge. Moenie net 'n antwoord neerskryf nie.* • *Stel duidelik alle aannames, reëls en wette wat u gebruik.* • *Waar van toepassing, teken die vryliggaamsdiagram (of kragte diagramme).* • *Wys dat jy die fisika wat jy gebruik om die vraag te beantwoord verstaan. Dit beteken dat volpunte sal nie noodwendig vir 'n numeries-korrekte antwoord gegee word nie.* • *Programmeerbare sakrekenaars MAG NIE gebruik word nie.* • *Geen handboeke word toegelaat nie.* • *'n Formuleblad is aangeheg.* • *Alle toets- en eksamenregulasies van die Universiteit van Pretoria is van toepassing.*

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Semester Test 1 | **FSK 116** | **Semestertoets 1**

15 March 2013 / *15 Maart 2013*

Surname and initials *Van en voorletters*		Student number *Studentenommer*	
Signature *Handtekening*		Venue *Lokaal*	

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Th 8:30 - 9:30
Fri 9:30 - 10:30

Question 1

a) Estimate (to an accuracy of 2 significant figures) the number of breaths taking during an average human lifetime. You can assume that a typical human lifetime is 70 years and the average number of breathes that a person takes is 10 breathes per minute. Clearly show all your units and conversions which you use during the calculation.

Skat (tot 'n akkuraatheid van 2 beduidende syfers) die aantal asemteue wat 'n gemiddelde mens in 'n leeftyd inasem. Neem aan dat 'n tipiese persoon 'n leeftyd het van 70 jaar, en 'n gemiddeld van 10 asemteue per minuut inasem. Wys duidelik al die eenhede en omskakelings wat jy in die berekening gebruik.

[4]

Number of minutes in a year: $1 \text{ year} = \frac{365 \text{ days}}{1 \text{ year}} \times \frac{24 \text{ hours}}{1 \text{ day}} \times \frac{60 \text{ minutes}}{1 \text{ hour}} = 5.26 \times 10^5 \text{ minutes}$ ✓✓
Number of breaths in a lifetime: $525.6 \times 10^3 \text{ minutes} \times 70 \text{ years} = 3.68 \times 10^7 \text{ minutes}$ ✓
Total number of breaths: $3.68 \times 10^7 \text{ minutes} \times \frac{10 \text{ breaths}}{1 \text{ minute}} = 3.68 \times 10^8 \text{ breaths}$ ✓
You need to show that the units cancel out – ie 365 days/year x 24 hours/day. Do not just write down numbers – we are not doing arithmetic, we are busy with Physics.

b) A burette of which the cross-sectional area, A, is constant, contains water to a height $h = h_0$. When the stopcock is opened the volume of water flows out at a rate of $A\frac{dh}{dt}$. If the stopcock is opened slightly, the height of the water in the tube is given by the function $h = h_0 e^{-\lambda t}$ metres in which the time, t, is measured in seconds and λ is a positive **constant** with unit s^{-1}. Determine the rate (volume per unit time) at which the water flows from the burette and explain the negative sign of the answer.

'n Buret met 'n konstante deursnee-oppervlakte, A, is gevul met water tot by 'n hoogte $h = h_0$. As die kraantjie oopgemaak word, is die volume water wat per tydseenheid uitvloei, $A\frac{dh}{dt}$. As die kraantjie baie klein oopgedraai word, word die hoogte van die water gegee deur die funksie $h = h_0 e^{-\lambda t}$ meter waar die tyd, t, in sekonde gemeet word en λ 'n positiewe ***konstante*** *is met eenheid s^{-1}. Bepaal die uitvloeitempo (volume per tydseenheid) van die water en verklaar die negatiewe teken van die antwoord.*

[4]

Flow rate $= A\frac{dh}{dt} = A\frac{d}{dt}\left(h_0 e^{-\lambda t}\right)$ ✓ $= Ah_0(-\lambda)e^{-\lambda t} = -Ah_0\lambda e^{-\lambda t}$ ✓✓
The negative sign indicates that height is decreasing with time. ✓
The flow rate is given as volume per unit time. $A\frac{dh}{dt}$ area x height = Adh = Volume

c) Calculate the area bounded by the t-axis and the graph of the function $v = 3 + 4t$, between t = 1s and t = 5s. The units of v are metres per second and that of t is seconds. What are the units of the quantity which the area represents?
Bereken die oppervlakte begrens tussen die t-as en die grafiek van die funksie $v = 3 + 4t$, tussen t = 1s en t = 5s. Die eenhede van v is meter per sekonde en dié van t is sekondes. Wat is die eenhede van die oppervlakte wat bereken is?

[4]

$v = 3 + 4t$

Area under the curve is the integral of v with respect to t.

$$\int v\,dt = \int (3 + 4t)\,dt$$

Between the limits t = 1 s and t = 5 s:

$$\int_1^5 v\,dt = \int_1^5 (3 + 4t)\,dt = \left[3t + 2t^2\right]_1^5$$

$$= \left[3(5) + 2(5)^2\right] - \left[3(1) + 2(1)^2\right]$$

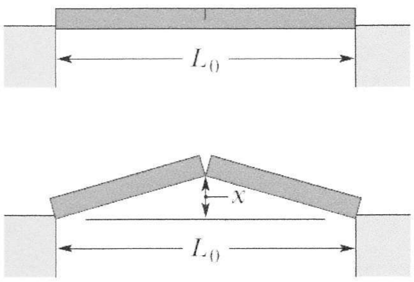
$$= [15 + 50] - [3 + 2] = 60$$

This represents a distance in metres : 60 m

The area under the curve requires integration with respect to t which is the variable along the horizontal axis of the graph. The units for v are m/s and that for the vaiable t is seconds.For the integral vdt, the units will be (m/s)(s) which is m.

Question 2

a) As a result of a temperature rise of ΔT, a bar with a crack at its centre buckles upward. If the fixed distance is L_0 and the coefficient of linear expansion of the bar is α:
'n Gekraakte staaf buig opwaarts in die middle as gevolg van 'n ΔT toename in die temperatuur. Indien die afstand L_0 vas is en die koëffisiënt van lineêre uitsetting is α:



- Find find the rise x of the centre.
  Bepaal die afstand x wat die middelpunt opwaarts beweeg.

[4]

Firstly determine the change in length of half the bar.

Add this to the original length of half the bar and this will determine the hypotenuse of the triangle.

Consider half the bar. Its original length is $\ell_0 = L_0 / 2$
and its length after the temperature increase is $\ell = L_0 + \alpha L_0 \Delta T$.
The old position of the half-bar, its new position, and the distance x that one end is displaced form a right triangle, with a hypotenuse of length ℓ, one side of length L_0, and the other side of length x.
The Pythagorean theorem yields

$$\ell^2 = \ell_0^2 + x^2 \quad ✓$$

$$x^2 = \ell^2 - \ell_0^2 = \ell_0^2(1+\alpha\Delta T)^2 - \ell_0^2. \quad ✓$$

The following approximation can be used, but not necessary.
Since the change in length is small, we may approximate $(1 + \alpha\Delta T)^2$ by $1 + 2\alpha\Delta T$, where the small term $(\alpha\Delta T)^2$ was neglected. Then,

$$x^2 = \ell_0^2 + 2\ell_0^2\alpha\Delta T - \ell_0^2 = 2\ell_0^2\alpha\Delta T$$

There are two approaches to this problem –
Option 1: You can divide the bar in two from the beginning and determine the change in length of half the bar. Add this to the original length of half the bar. You now have two sides of a right-angled triangle – so it is then easy to determine the third side.
Option 2: Determine the change in length of the original bar, determine the final length of the whole bar. Then you determine half the length of the original and the elongated bars, and again you have 2 sides of the right-angled triangle.
No values are used in this part of the question. Only symbols are used.

- If the temperature rise is 32°C, the distance L_0 = 3.77 m and the coefficient of linear expansion of the bar is 25 × 10^{-6}/C°, what is the value of x?
  Indien die tempertuurverandering 32°C is, die afstand L_0 = 3.77 m en die koëffisiënt van lineêre uitsetting is 25 × 10^{-6}/C°, wat is die waarde van x?

[2]

$$x^2 = \ell_0^2(1+\alpha\Delta T)^2 - \ell_0^2.$$

$$= \left(\frac{3.77}{2}\text{ m}\right)^2\left(1+\left(25\times10^{-6}/\text{C}°\right)\left(32°\text{ C}\right)\right)^2 - \left(\frac{3.77}{2}\text{ m}\right)^2 \quad ✓$$

$$= 5.69\times10^{-3}\text{ m}^2$$

$$x = 7.54\times10^{-2}\text{ m} \quad ✓$$

From the approximation we get the same answer:

$$x = \ell_0\sqrt{2\alpha\Delta T} = \frac{3.77\text{ m}}{2}\sqrt{2(25\times10^{-6}/\text{C}°)(32°\text{C})} = 7.5\times10^{-2}\text{ m}.$$

Use the result that you obtained above and calculate the value of x.

b) The temperature of a 0.700 kg cube of ice is found to be –150 °C. Energy is gradually transferred to the cube as heat while it is otherwise thermally isolated from its environment. The total heat transferred is 0.6993 MJ. Assume that the specific heat of ice, c = 2220 J/kg·K is valid for the all temperatures from –150°C to 0°C. What is the final state of the H_2O and what is the final temperature? L_f = 333 kJ/kg, L_v = 2256 kJ/kg, c_{water} = 4187 J/kg·K

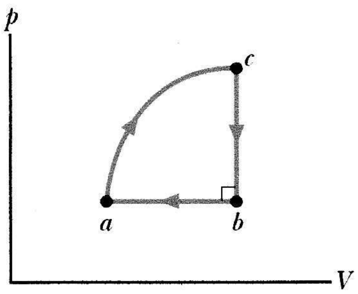
Die temperatuur van 'n 0.700 kg blok ys is –150°C. Energie in die vorm van warmte word gelydelik oorgedra aan die blok wat termies geïsoleer is van die omgewing. Die totale hoeveelheid warmte oorgedra is 0.6993 MJ. Veronderstel dat die spesifieke warmte van ys, c = 2220 J/kg·K, is geldig vir alle temperature vanaf –150°C tot 0°C. Wat is die finale toestand van die H_2O en wat is die finale temperatuur? L_f = 333 kJ/kg, L_v = 2256 kJ/kg, c_{water} = 4187 J/kg·K.

[7]

Here we have a certain amount of energy available and we need to determine the state and temperature of the system when this energy has been absorbed. Explain each step you take in this calculation. Determine first of all if the ice is going to be able to reach 0°C. If you still have energy available, then determine if the ice will melt. If you still have energy available, determine the maximum temperature that will be reached by the water. If this is above 100°C, then determine how much of the water will vapourize.
Total energy in the form of heat available is 0.6993 MJ = 6.993 × 10^5 J
To get the ice from –150°C to 0°C: $Q = mc_{ice}\Delta T = 0.700\text{ kg}\left(2220\text{ J/kg}\cdot\text{K}\right)\left(150\text{ K}\right) = 2.331\times10^5\text{ J}$ ✓ ✓
There is still energy available To melt the ice at 0°C: $Q = mL_f = 0.700\text{ kg}\left(333\times10^3\text{ J/kg}\cdot\text{K}\right) = 2.331\times10^5\text{ J}$ ✓ ✓
Total energy used: 4.662 × 10^5 J Energy still available: 2.332 × 10^5 J to raise the temperature. ✓ $Q = mc_{ice}\Delta T$ $\Delta T = \dfrac{Q}{mc_{water}} = \dfrac{2.331\times10^5\text{ J}}{0.700\text{ kg}\left(4187\text{ J/kg}\cdot\text{K}\right)} = 79.5°\text{C}$ ✓ ✓ Approx 80°C will do

5,8, 19, 28, 31, 49, 50, 57, 65

c) The figure displays a closed cycle for a gas. For c to b, 40 J is transferred from the gas as heat. From b to a, 130 J is transferred from the gas as heat, and the magnitude of the work done on the gas is 80 J. From a to c, 400 J is transferred to the gas as heat. What is the work done by the gas from a to c? Clearly show all your reasoning.
Die figuur toon 'n geslote siklus vir 'n gas aan. Van c tot b word 40 J oorgedra vanuit die gas as warmte. Van b tot a word 130 J oorgedra vanuit die gas as warmte, en die hoeveelheid arbeid gedoen op die gas is 80 J. Van a tot c word 400 J oorgedra na die gas as warmte. Hoeveel arbeid word gedoen deur die gas van a tot c? Wys duidelik met redes hoe jy jou die antwoord bereken.



[5]

For each section of this closed cycle, Q and $\Delta E = Q - W$

For the whole cycle we can add up all the ΔE for each section and they must add up to 0J.

From $c-b$: Q = -40 J	W = 0 J (No volume change)	$\Delta E_{cb} = Q - W = -40 - 0 = -40$ J ✓
From $c-b$: Q = -130 J	W = 80 J (No volume change)	$\Delta E_{ba} = Q - W = -130 - 80$ $= -210$ J ✓

For a closed cycle: $\Delta E = 0$ J for *c-b-a-c*

$\Delta E_{cbac} = \Delta E_{cb} + \Delta E_{ba} + \Delta E_{ac} = 0\text{J}$ ✓

$\Delta E_{ac} = -(\Delta E_{cb} + \Delta E_{ba}) = -(-40 + (-210))$ J = 250 J for ✓ *c-b-a-c*

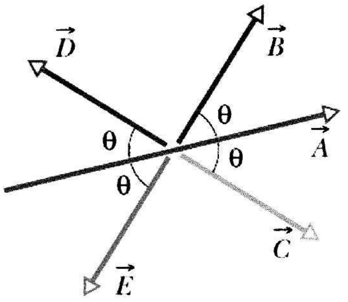
From $a-c$: Q = 400 J

$$\Delta E = Q - W$$
$$W = Q - \Delta E = 400 - 250$$
$$= 150 \text{ J}$$ ✓

The work must be positive since the volume is increasing.

Question 3

a) The figure shows vector $\vec{A}$, and the four vectors $\vec{B}$, $\vec{C}$, $\vec{D}$ and $\vec{E}$, that have the same magnitude but differ in orientation. Which of the four vectors $\vec{B}$, $\vec{C}$, $\vec{D}$ and $\vec{E}$ have the same dot product with $\vec{A}$?
Die figuur vertoon vektor $\vec{A}$, en vier ander vektore $\vec{B}$, $\vec{C}$, $\vec{D}$ en $\vec{E}$, wat dieselfe grootte maar verskillende oriëntasies het. Watter van die vier vektore $\vec{B}$, $\vec{C}$, $\vec{D}$ en $\vec{E}$ het dieselfde puntproduk met $\vec{A}$?

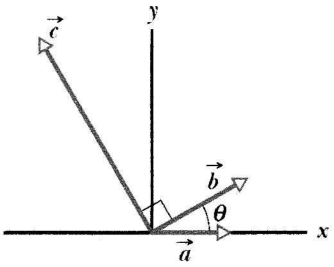


[4]

The dot products of $\vec{A}\cdot\vec{B}$ and $\vec{A}\cdot\vec{C}$ have the same value. ✓✓
The dot products of $\vec{A}\cdot\vec{D}$ and $\vec{A}\cdot\vec{E}$ have the same value. ✓✓ These values will be negative.

b) The three vectors in the Figure have magnitudes a = 3.00 m, b = 4.00 m and c = 10.0 m. The angle θ = 30.0°.
Die drie vektore in die figuur het groottes a = 3.00 m, b = 4.00 m en c = 10.0 m. Die hoek θ = 30.0°.

- Write these vectors in unit vector notation.
Skryf die vektore in eenheidsvektor-notasie.



[6]

$\vec{a}=\left(3.00\ \hat{\imath}\right)\text{m}$ ✓✓
$\vec{b}=\left(4.00\cos 30.0°\hat{\imath}+4.0\sin 30.0°\hat{\jmath}\right)\text{m}=\left(3.46\hat{\imath}+2.00\hat{\jmath}\right)\text{m}$ ✓✓
$\vec{c}=\left(10.0\cos 120°\hat{\imath}+10.0\sin 120°\hat{\jmath}\right)\text{m}=\left(-5.00\hat{\imath}+8.66\hat{\jmath}\right)\text{m}$ ✓✓

- Determine the magnitude and direction of the cross product of $\vec{b}\times\vec{c}$.
Bepaal die grootte en rigting van die kruisproduk $\vec{b}\times\vec{c}$.

[3]

The cross product: $\vec{b}\times\vec{c}=bc\sin\theta=(4)(10)\sin 90°=40\ \text{m}^2$ ✓ ✓ do not forget the units
The direction of the cross-product vector will be out of the page. ✓

Question 4

a) When the acceleration is **constant**, the average and instantaneous acceleration are equal: $a = a_{avg} = \dfrac{v - v_0}{t - 0}$, where a is the acceleration, v is the velocity at time t and v_0 is the initial velocity at $t = 0$ s. We can also write the average velocity as $v_{avg} = \dfrac{x - x_0}{t - 0}$, where x is the position at time t and x_0 is the initial position at $t = 0$ s. Using these two equations (**not the integration method**) show that when a particle is accelerating in the x direction, its velocity is given by $v^2 = v_0^2 + 2a(x - x_0)$. Show all steps and state all reasoning clearly.

Wanneer die versnelling ***konstant*** *is, is die gemiddelde en oombliklike versnelling dieselfde:* $a = a_{gem} = \dfrac{v - v_0}{t - 0}$ *waar a die versnelling is, v is die snelheid by tyd t en* v_0 *is die aanvanklike snelheid by* $t = 0$ *s. Ons kan ook die gemiddelde snelheid skryf as* $v_{gem} = \dfrac{x - x_0}{t - 0}$, *waar x die posisie by tyd t is en* x_0 *die aanvanklike posisie by* $t = 0$ *s is. Deur gebruik te maak van dié twee vergelykings (****nie die integrasie-metode nie****), bewys dat die snelheid gegee word deur* $v^2 = v_0^2 + 2a(x - x_0)$. *Wys al die stappe en verduidelik jou aannames.*

[9]

$$a = a_{avg} = \frac{v - v_0}{t - 0} \qquad ① \checkmark$$

$$v = v_0 + at$$

$$v_{avg} = \frac{x - x_0}{t - 0} \qquad ② \checkmark$$

$$x = x_0 + v_{avg}t$$

But $v_{avg} = ½(v + v_0)$. ✓

Substituting this for v_{avg} in ② :

$x = x_0 + [½(v + v_0)]t$ ✓

From the first equation, we substitute for v and rearrange the terms:

$$x = x_0 + [½(v + v_0)]t \quad \checkmark$$

$$\begin{aligned} x - x_0 &= ½vt + ½v_0t \\ &= ½(v_0 + at)t + ½v_0t \\ &= v_0t + ½at^2 \end{aligned}$$

From ①: $t = \dfrac{v - v_0}{a}$ ✓

Substituting into above:

✓

$$x - x_0 = v_0 t + \tfrac{1}{2}at^2 \quad ✓$$

$$= v_0\left(\frac{v - v_0}{a}\right) + \tfrac{1}{2}a\left(\frac{v - v_0}{a}\right)^2$$

$$2(x - x_0)a = 2v_0(v - v_0) + (v - v_0)^2$$

$$2a\Delta x = 2v_0 v - 2{v_0}^2 + v^2 + {v_0}^2 - 2v_0 v \quad ✓$$

$$v^2 = {v_0}^2 + 2a(x - x_0) \quad ✓$$

Shorter method

$$a = a_{avg} = \frac{v - v_0}{t - 0} \qquad ① \quad ✓✓$$

$$v_{avg} = \frac{x - x_0}{t - 0} \qquad ② \quad ✓$$

$$x = x_0 + v_{avg} t$$

From ①: $t = \dfrac{v - v_0}{a}$ ③ ✓✓

Substituting ③ in ②:

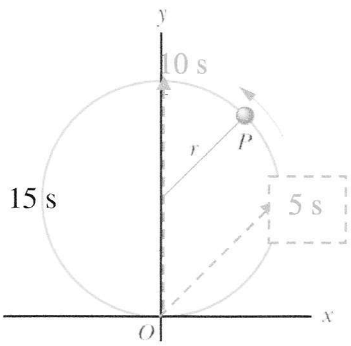
$$\frac{(x - x_0)a}{v - v_0} = \tfrac{1}{2}(v + v_0) \quad ✓✓$$

$$2(x - x_0)a = (v + v_0)(v - v_0) \quad ✓$$

$$= v^2 - v_0^2 \quad ✓$$

$$v^2 = v_0^2 + 2a(x - x_0)$$

b) A particle P travels with constant speed on a circle of radius $r = 3.00$ m and completes one revolution in 20.0 s. The particle passes through O at time $t = 0$ s. The following vectors must be given in terms of magnitude and their direction from the positive x-axis.
'n Deeltjie P beweeg met konstante spoed op 'n sirkel met radius r = 3.00 m en dit voltooi een omwenteling in 20.0 s. Die deeltjie gaan deur O wanneer t = 0 s. Die volgende vektore moet in terme van die grootte en rigting ten opsigte van die positiewe x-as gegee word.



- The position vector at $t = 5.00$ s.
  Die posisie-vektor by t = 5.00 s.

[2]

After 5 s the particle has move ¼ around the circular path. Its position will be $\vec{r} = \left(3.00\hat{\imath} + 3.00\hat{\jmath}\right)$ m

The magnitude of this position vector is: $|\vec{r}| = \sqrt{r_x^2 + r_y^2} = \sqrt{18} = 4.24$ m ✓

At an angle of 45° above the positive x-axis ✓

- The position vector at t = 10.0 s.
  Die posisie vector by t = 10.0 s.

[2]

After 10 s the particle has move ½ way around the circular path. Its position will be $\vec{r} = \left(0.00\hat{\mathrm{i}} + 6.00\hat{\mathrm{j}}\right)\mathrm{m}$
The magnitude of this position vector is: $\left\|\vec{r}\right\| = \sqrt{r_x^2 + r_y^2} = 6.00\ \mathrm{m}$ ✓ At an angle of 90° anticlockwise from the positive x-axis. ✓

- The displacement of the particle during the time t = 5.00 s and t = 10.0 s.
  Die verplasing van die deeltjie in die tydsinterval t = 5.00 s en t = 10.0 s.

[2]

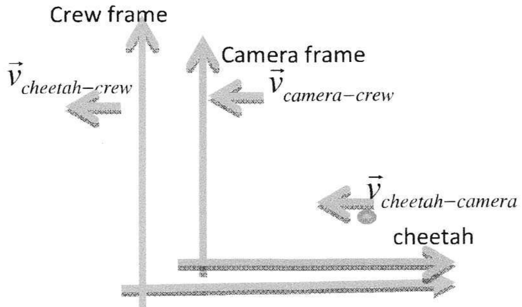
$\Delta\vec{r} = \vec{r}_{10s} - \vec{r}_{5s} = \left[\left(0.00\hat{\mathrm{i}} + 6.00\hat{\mathrm{j}}\right)\mathrm{m} - \left(3.00\hat{\mathrm{i}} + 3.00\hat{\mathrm{j}}\right)\mathrm{m}\right]$ $= \left(-3.00\hat{\mathrm{i}} + 3.00\hat{\mathrm{j}}\right)\mathrm{m}$
$\left\|\Delta\vec{r}\right\| = \sqrt{\Delta r_x^2 + \Delta r_y^2} = 4.24\ \mathrm{m}$ ✓ Direction is at an angle of (45°+90° =) 135° anticlockwise from the positive x-axis ✓

- The acceleration at t = 5.00 s.
  Die versnelling by t = 5.00 s

[3]

Speed of the particle can be determined from: $v = \dfrac{\text{distance}}{time} = \dfrac{2\pi r}{\Delta t}$ ✓
$a = \dfrac{v^2}{r} = \left(\dfrac{2\pi r}{\Delta t}\right)^2 \dfrac{1}{r} = \dfrac{4\pi^2 r}{(\Delta t)^2}$ ✓ $= \dfrac{4\pi^2 (3.00)}{20\mathrm{s}}$ ✓ $= 0.296\ \mathrm{m/s^2}$
Directed to the centre of the circular path.

c) A cameraman on a pickup truck is traveling westward at 20 km/h while he records a cheetah that is moving westward 30 km/h faster than the truck. Suddenly, the cheetah stops, turns, and then runs at 45km/h eastward, as measured by a suddenly nervous crew member hiding behind a bush alongside the cheetah's path. The change in the animal's velocity takes 2.0 s.
'n Kameraman staan agter op 'n bakkie wat in 'n westelike rigting beweeg teen 20 km/h terwyl hy 'n jagluiperd opneem wat ook in 'n westelike rigting beweeg teen 30 km/h vinniger as die bakkie. Die jagluiperd stop skielik, draai, en hardloop dan in 'n oostelike rigting teen 45 km/h soos gemeet deur 'n baie senuweeagtige spanlid wat agter 'n bos wegkruip langs die jagluiperd se pad. Die verandering in die dier se snelheid neem 2.0 s.



This is a diagram to illustrate the two reference frames which are applicable here. The red one is the reference frame of the crew (ground) and the blue one is the cameraman on the truck moving at a constant velocity.

- What are the magnitude and direction of the animal's acceleration according to the cameraman?
  Wat is die grootte en rigting van die dier se versnelling volgens die kameraman?

[5]

$$x_{cheetah-crew} = x_{cheetah-camera} + x_{camera-crew}$$
$$\vec{v}_{cheetah-crew} = \vec{v}_{cheetah-camera} + \vec{v}_{camera-crew}$$

Initially:

$$\vec{v}_{cheetah-crew} = \vec{v}_{cheetah-camera} + \vec{v}_{camera-crew}$$
$$= -30 \text{ km/h} + (-20 \text{ km/h})$$
$$= -50 \text{ km/h}$$

$$\vec{v}_{cheetah-camera} = -30 \text{ km/h} \times \frac{1000 \text{ m}}{\text{km}} \times \frac{\text{h}}{3600 \text{ s}}$$
$$= -8.33 \text{ m/s}$$

Finally:

$$\vec{v}_{cheetah-crew} = \vec{v}_{cheetah-camera} + \vec{v}_{camera-crew}$$
$$\vec{v}_{cheetah-camera} = \vec{v}_{cheetah-crew} - \vec{v}_{camera-crew}$$
$$= +45 \text{ km/h} - (-20 \text{ km/h})$$
$$= 65 \text{ km/h}$$

Acceleration of cheetah as measured by the cameraman

$$a = \frac{\Delta\vec{v}}{\Delta t} = \frac{18.1 \text{ m/s} - (-8.3 \text{ m/s})}{20 \text{ s}} = 13.2 \text{ m/s}^2 \text{ in an easterly direction}$$

- What are the magnitude and direction of the animal's acceleration according to the nervous crew member?
  Wat is die grootte en rigting van die dier se versnelling volgens die senuweeagtige spanlid?

[2]

No calculation is actually needed here. The two reference frames are not accelerating with respect to each other, and therefore they should measure the same acceleration of the cheetah.
Since the truck with the cameraman is not accelerating the acceleration will be the same for both reference frames. Therefore the acceleration as measured by the crew member is also $a = 13.2\ \text{m/s}^2$ ✓ in an easterly direction ✓
Alternate method: Initially - $\vec{v}_{cheetah-camera} = 50\ \text{km/h} \times \frac{1000\ \text{m}}{\text{km}} \times \frac{\text{h}}{3600\ \text{s}} = 13.8\ \text{m/s}$ Finally - $\vec{v}_{cheetah-camera} = 45\ \text{km/h} \times \frac{1000\ \text{m}}{\text{km}} \times \frac{\text{h}}{3600\ \text{s}} = 12.5\ \text{m/s}$
$a = \frac{\Delta\vec{v}}{\Delta t} = \frac{12.5\ \text{m/s} - (-13.8\ \text{m/s})}{2\ \text{s}} = 13.2\ \text{m/s}^2$ ✓ in an easterly direction ✓

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